UWB Anchor Self-Localization in 3D: Algorithm Comparison under Noise, Link Loss and NLOS

Bc. Inessa Davtyan
Faculty of Electrical Engineering, Czech Technical University in Prague
October 2026

Abstract — Ultra-wideband (UWB) positioning needs known anchor coordinates, but only the pairwise two-way ranges between anchors are measured. This research compares four families of range-only 3D self-localization algorithms in Monte Carlo simulation against the Cramér–Rao bound (CRB). Nonlinear least squares (NLS) initialized by classical MDS reaches the CRB on clean data. Under link loss, a multi-start variant cuts gross failures from 30 % to 3 % at 20 % missing links. A soft-ℓ_1 loss lowers the median error by 27–33 % at 5–10 % NLOS links. Anchor sets with ≤ 2 m height spread have a vertical error 5 – 6 times the horizontal one. The covariance $\sigma^2(\mathbf{J}^\top\mathbf{J})^+$ is calibrated: its 95 % ellipse covers 94 % of 400 Monte Carlo estimates.

1. Introduction

Surveying anchors by hand is slow, error-prone and must be repeated whenever the constellation changes [2]. Self-localization estimates anchor coordinates from the $N(N-1)/2$ inter-anchor ranges alone. The result is a relative map: ranges fix the shape up to a rigid motion and a reflection (Fig. 1b), so an absolute frame needs additional reference points. This research asks which algorithm is accurate, robust to missing and biased links, cheap and able to report its own uncertainty. Contributions:

- a literature overview and a unified formulation with an analytical bound,
- a paired Monte Carlo comparison of six variants over noise, anchor count, link loss, NLOS and height (Figs. 3–10),
- a calibrated covariance and a recommended pipeline (Fig. 2, Table 5).

2. Problem and Background

2.1 Model, objective, bound

For anchors $x_1, \ldots, x_N \in \mathbb{R}^3$ and a measured link set $\mathcal{E}$,

$$d_{ij} = \|x_i - x_j\| + n_{ij} + b_{ij}, \quad n_{ij} \sim \mathcal{N}(0, \sigma^2),\ b_{ij} \geq 0, \tag{1}$$

$$\hat{X} = \arg\min_X \sum_{(i,j)\in\mathcal{E}} \rho(\|x_i - x_j\| - d_{ij}), \tag{2}$$

with $\rho(r) = r^2$ (least squares) or the soft-ℓ_1 loss $\rho(r) = 2f^2(\sqrt{1+(r/f)^2} - 1)$, $f = 0.3$ m. With $u_{ij} = (x_i - x_j)/\|x_i - x_j\|$ the Fisher information is

$$\mathbf{J}_F = \sigma^{-2} \sum_{(i,j)\in\mathcal{E}} (e_i - e_j)(e_i - e_j)^\top \otimes u_{ij}u_{ij}^\top,$$

$$\text{RMSE} \geq \sqrt{\text{tr}(\mathbf{J}_F^+)/N}. \tag{3}$$

The pseudo-inverse removes the six rigid-motion modes. Unique realizability needs a (globally) rigid graph with $\geq 3N-6$ links and at least four per node [3, 6]. Otherwise noise selects among flip solutions [7].

3. Methods

cMDS: fill missing ranges by shortest paths, double-center the squared matrix, take the three largest eigenpairs.

SMACOF: weighted stress majorization from the cMDS start.

NLS/LM: trust-region Levenberg–Marquardt on (2) with the analytic Jacobian, from the cMDS start. **Multi-start** adds seven random starts and keeps the lowest cost. **Soft-ℓ_1** adds the robust loss.

SDP: anchor-free Gram-matrix relaxation (CVXPY/Clarabel), rank-3 projection, trace weight not tuned. Treated as an offline reference.

Table 1. Compared estimators.

Estimator	Principle	Missing links	Cost
cMDS	closed form, N^3	shortest paths	1 EVD
SMACOF	majorization	weights	iterative
NLS/LM	Gauss–Newton family	only measured links	iterative
Multi-start (+ soft-ℓ_1)	8 NLS runs, min. cost	only measured links	8× NLS
SDP	convex relaxation	only measured links	interior point

3.1 Experimental protocol

Table 2. Simulation settings.

Volume	uniform in $20 \times 20 \times 6$ m, min. spacing 1 m
Noise / defaults	$\sigma = 10$ cm, $N = 8$, full mesh
Link loss	random removal, resampled until all nodes have ≥ 4 links and the framework is infinitesimally rigid
NLOS	fraction of links with bias $b \sim \mathcal{U}(0.5, 2)$ m
Trials	60 per sweep point (40 in heatmaps), paired: all methods share data
Metric	RMSE after Procrustes alignment, gross failure = RMSE > 1 m

Table 2 lists the settings. Runtimes are desktop CPU, Python, and indicative only.

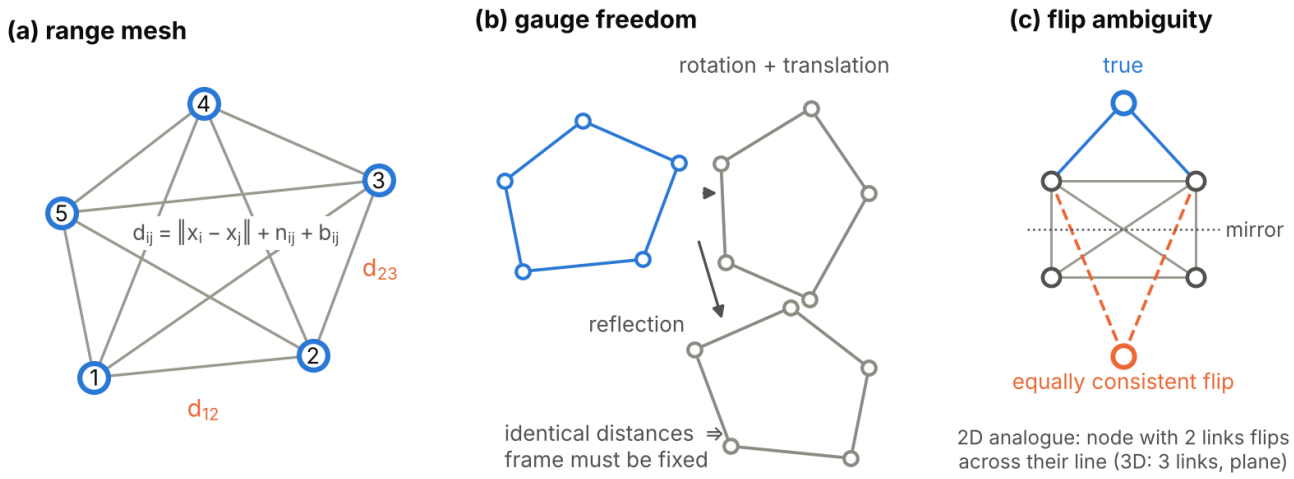


Figure 1. Problem structure. (a) Ranges d_{ij} between all anchor pairs form the only input. (b) They are invariant under rotation, translation and reflection, so a frame must be fixed externally. (c) Nodes with too few links can flip to a mirror position that explains all ranges equally well.

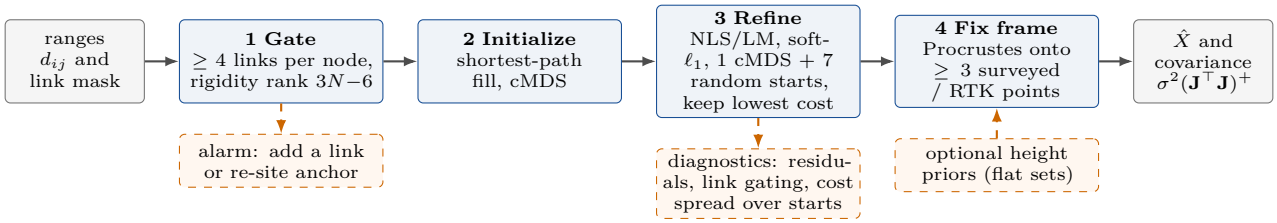


Figure 2. Recommended pipeline. Solid blocks are evaluated here. Dashed blocks are operational add-ons.

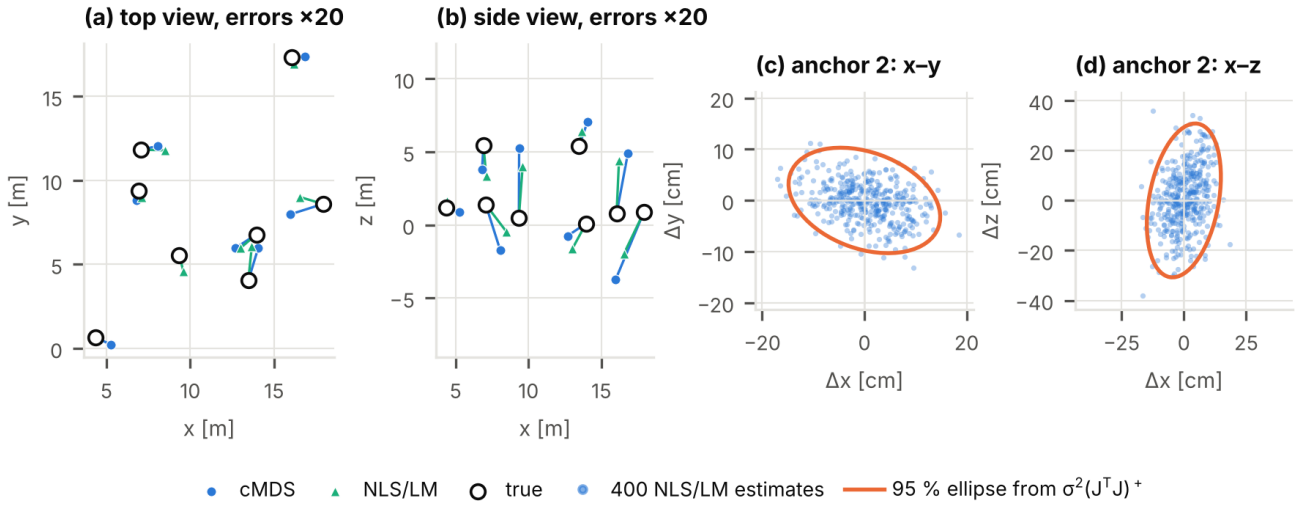


Figure 3. One constellation, many trials. (a, b) Estimates of cMDS and NLS/LM for one noisy mesh (N=8), errors magnified 20×. (c, d) 400 repeated noisy meshes of the same layout: scatter of the NLS/LM error of anchor 2 and the 95 % ellipse predicted by $\sigma^2(\mathbf{J}^\top\mathbf{J})^+$. The ellipse contains 94 % of the estimates. The vertical spread ($1\sigma \approx 13$ cm) is twice the horizontal one (6 cm).

4. Results

Key findings.

1. NLS/LM reaches the CRB on clean data. cMDS is 17–30 % above it (Fig. 4).
2. The CRB-based covariance is calibrated (94 % coverage, Fig. 3).
3. Multi-start removes gross failures up to 30 % link loss. The final cost identifies the flip (Figs. 8–9, 6).
4. Soft-ℓ_1 helps at 5–10 % NLOS only (Fig. 8c,d).
5. Vertical error dominates unless the anchors span ≥ 4 m in height (Fig. 10).
6. Cost grows 12–16× from single to multi-start (Fig. 7).

Table 3. Median RMSE vs. noise (N=8, cm). NLS/LM stays within ±9 % of the CRB. cMDS is 13–27 % above it.

σ	CRB	NLS/LM	cMDS	SDP
2	3.3	3.0	3.7	3.9
5	8.3	7.5	9.6	10.5
10	16.2	17.0	19.9	20.9
15	24.8	22.5	29.3	29.9
20	36.9	35.5	44.6	46.1
30	50.7	51.6	64.6	64.2

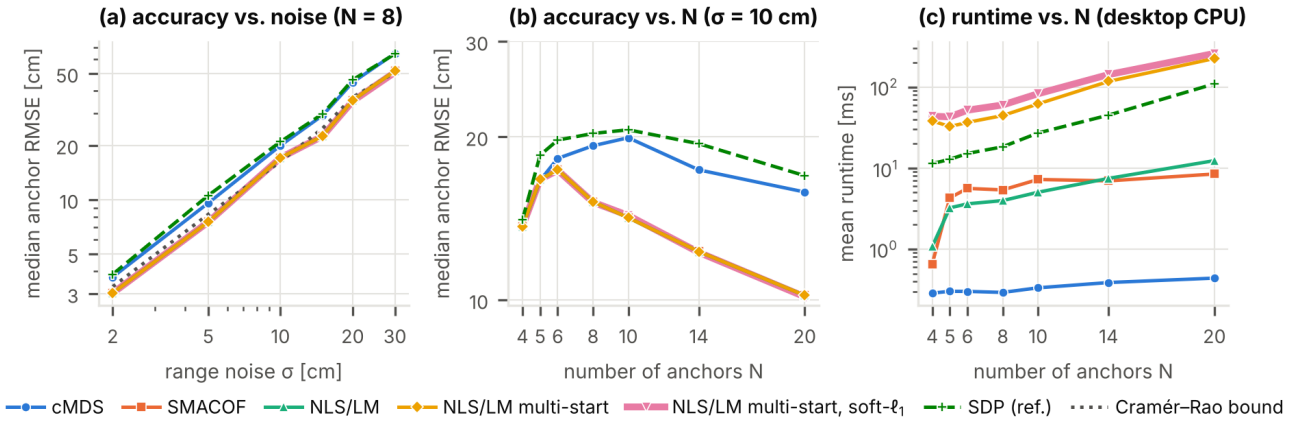


Figure 4. Clean-data accuracy and cost. Median over 60 trials. NLS/LM and SMACOF follow the CRB. cMDS and SDP are 17–30 % above it. More anchors help NLS (14 cm at N=10 → 10 cm at N=20) but not cMDS. Cost: 0.3 ms (cMDS), 1–12 ms (NLS), 30–260 ms (multi-start).

Table 4. Summary. Clean: N=8, σ=10 cm. Stress: N=12, 20 % missing links and 10 % NLOS links. Failure = RMSE > 1 m.

	clean		20 % links missing		10 % NLOS links		runtime
Method	median [cm]	p90 [cm]	median [cm]	failures [%]	median [cm]	failures [%]	N=10 [ms]
cMDS	19.9	31.4	302	100	68	25	0.3
SMACOF	17.0	28.7	47	33	63	23	7.2
NLS/LM (cMDS init)	17.0	28.7	23	30	63	23	5.0
NLS/LM multi-start	17.0	28.7	20	3	63	20	61.8
NLS/LM multi-start, soft-ℓ_1	17.0	28.9	20	2	46	17	82.6
SDP (reference)	20.9	33.1	27	2	56	20	27.1
Cramér–Rao bound	16.2						

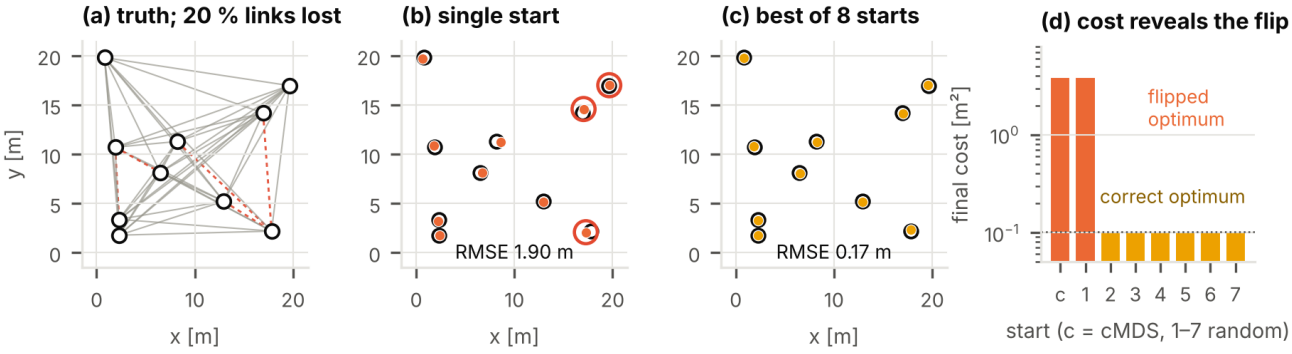


Figure 5. Anatomy of a gross failure (N=10, 20 % links lost, one trial). (a) Truth and lost links (red dashed). (b) Single-start NLS/LM from the cMDS start converges to a flipped configuration with three anchors off (red circles). (c) The best of eight starts is correct. (d) The final cost separates the two optima by a factor of 38, so the lowest-cost start is a reliable selector here.

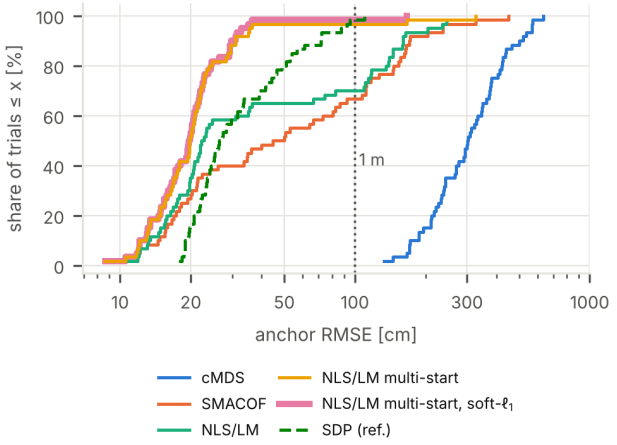


Figure 6. Distribution of errors at 20 % link loss (N=12). The multi-start curves rise steeply at 20 cm. Single-start methods have a heavy tail beyond 1 m.

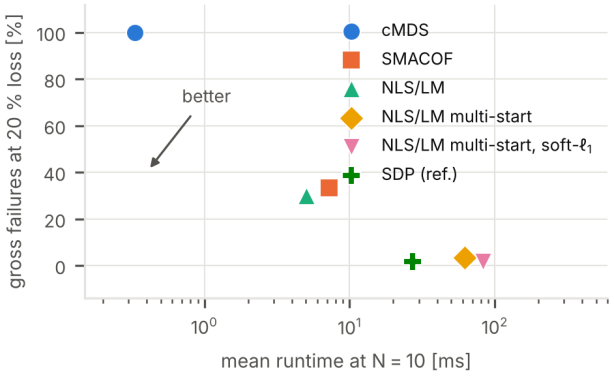


Figure 7. Robustness vs. cost. Multi-start variants and SDP sit in the low-failure region. Multi-start NLS costs ∼12–16× a single NLS run (desktop CPU, N=10).

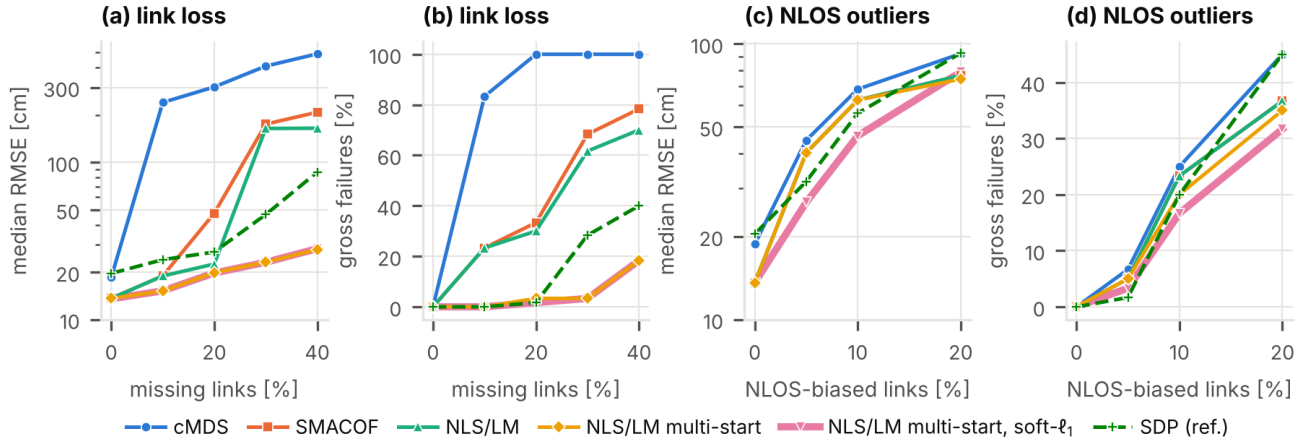


Figure 8. Missing links and NLOS (N=12, 60 trials). cMDS fails from 10 % link loss on because shortest-path fill inflates distances. Multi-start keeps failures ≤ 3 % up to 30 % loss. The soft-ℓ_1 loss helps at 5–10 % NLOS (27 vs. 40 cm at 5 %) and not at 20 %, where no method works.

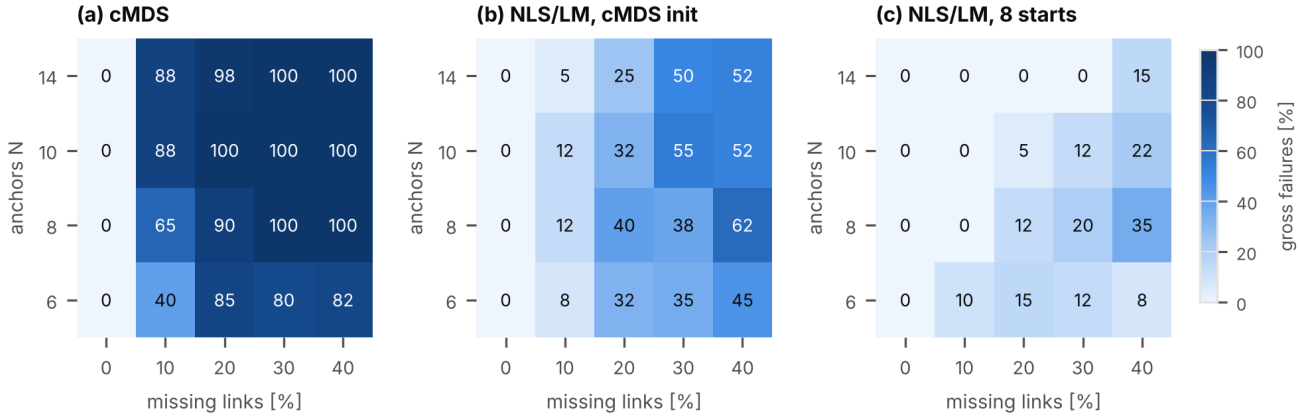


Figure 9. Failure rate over anchor count and link loss (40 trials per cell). Single-start refinement fails in 25–55 % of the meshes at 20–30 % loss for every N. Eight starts remove most failures. Binomial s.e. is up to 8 pp per cell.

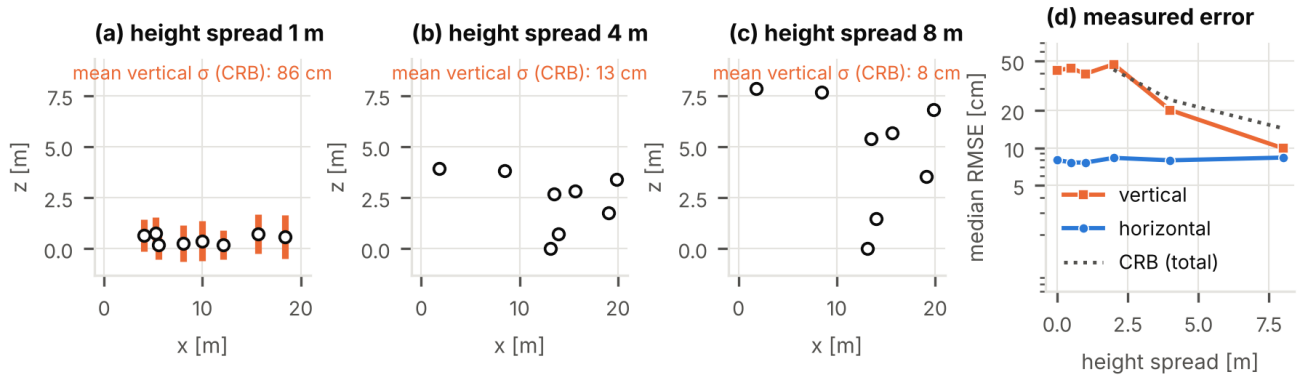


Figure 10. Anchor height spread (N=8). (a–c) Side views of three constellations. The bar length would be the CRB vertical standard deviation (mean value printed). (d) Measured median error of the NLS multi-start soft-ℓ_1 estimator: horizontal error stays at 8 cm. Vertical error is 5–6× larger up to 2 m height spread, 2.5× at 4 m and 1.2× at 8 m. The CRB (total) is shown only where it is well conditioned.

Link loss. At 20 % loss the cMDS start is wrong in every trial, so single-start refinement inherits a flipped or folded configuration in 30 % of the meshes (Fig. 6). The lowest-cost start rescues almost all of them (Fig. 5d). Even eight starts fail in 8–15 % of the meshes at N=6 and in up to 35 % at 40 % loss (Fig. 9).

Cost. Multi-start multiplies the runtime by the number of starts. Since random starts converge to the same optimum in clean data, an early exit once two starts agree would recover most of the speed.

5. Discussion

NLS/LM over all ranges, cMDS-initialized, soft-ℓ_1 loss, multi-start when the mesh is incomplete (Fig. 2). It reaches the CRB, is the most robust option tested at moderate link loss, and gives a covariance for free. Why not SDP? Comparable robustness to link loss but 23–40 % less accurate here (trace weight untuned), ∼5× slower than a single NLS. Procrustes onto ≥ 3 surveyed points gives an absolute frame. Reference quality then adds directly to the error budget.

Table 5. Decision matrix. Shading: green = good, yellow = acceptable, red = poor. Grey = literature only, not tested.

Method	Clean accuracy	Missing links	NLOS (10 %)	Compute	Covariance
cMDS	17–30 % above NLS	83 % fail at 10 % loss	68 cm	0.3 ms	none
SMACOF	at CRB	33 % fail at 20 %	63 cm	7 ms	none
NLS/LM, cMDS init	at CRB	30 % fail at 20 %	63 cm	5 ms	$\sigma^2(\mathbf{J}^\top\mathbf{J})^+$
NLS/LM multi-start, soft-ℓ_1	at CRB	2 % fail at 20 %	46 cm	83 ms	$\sigma^2(\mathbf{J}^\top\mathbf{J})^+$
SDP	23–40 % above NLS	2 % at 20 %, 28 % at 30 %	56 cm	27 ms	none
Grid / particle filter [18, 19]	$<$ 15 cm reported	reported robust	reported robust	heavy	posterior

Limitations. Simulation only. Gaussian noise and uniform positive NLOS bias, no antenna-delay error. Random volumes. Runtime not measured on embedded hardware. Multi-start cannot detect a consistent flip, and soft-ℓ_1 helps little above 10 % NLOS. Height priors and dynamic anchor counts were not simulated.

Table 6. Practical guidelines derived from the experiments.

Factor	Evidence	Guideline
Links	single start fails 25–55 % at 20–30 % loss	≥ 4 links per node, rigid, use multi-start below full mesh.
Height	vertical error 5–6× horizontal at ≤ 2 m	Spread anchors over ≥ 4 m or add measured height priors.
NLOS	soft-ℓ_1 gains 27–33 % at 5–10 %	Gate suspicious links, do not expect help above 10 %.
Count	14 cm (N=10) → 10 cm (N=20)	More anchors reduce error, 8 is a workable minimum.
Frame	Procrustes needs ≥ 3 non-collinear points	Survey or RTK three anchors.

6. Conclusion

Range-only 3D anchor self-localization is best solved by robust, multi-start NLS refinement of an MDS initial guess. It matches the CRB, survives 20–30 % link loss and cuts the damage of moderate NLOS by about 30 %, at tens of milliseconds on a desktop CPU. The dominant residual risks are flat constellations (vertical error) and heavy NLOS contamination. Both call for geometry design and link gating rather than a different estimator.

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